

CLINICAL AND POPULATION SCIENCES

Can Children With Perinatal Stroke Use a Simple Brain Computer Interface?

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BACKGROUND AND PURPOSE: Perinatal stroke is the leading cause of hemiparetic cerebral palsy resulting in lifelong disability for millions of people worldwide. Options for motor rehabilitation are limited, especially for the most severely affected children. Brain computer interfaces (BCIs) sample brain activity to allow users to control external devices. Functional electrical stimulation enhances motor recovery after stroke, and BCI-activated functional electrical stimulation was recently shown to improve upper extremity function in adult stroke. We aimed to determine the ability of children with perinatal stroke to operate a simple BCI.

METHODS: Twenty-one children with magnetic resonance imaging–confirmed perinatal stroke (57% male, mean [SD] 13.5 [2.6] years, range 9–18) were compared with 24 typically developing controls (71% male, mean age [SD] 13.7 [3.7] years, range 6–18). Participants trained on a simple EEG-based BCI over 2 sessions (10 trials each) utilizing 2 different mental imagery strategies: (1) motor imagery (imagine opening and closing of hands) and (2) goal oriented (imagine effector object moving toward target) to complete 2 tasks: (1) drive a remote controlled car to a target and (2) move a computer cursor to a target. Primary outcome was Cohen Kappa with a score >0.40 suggesting BCI competence.

RESULTS: BCI performance was comparable between stroke and control participants. Mean scores were 0.39 (0.18) for stroke versus 0.42 (0.18) for controls ($t[42]=0.478$, $P=0.94$). No difference in performance between venous ($M=0.45$, $SD=0.29$) and arterial ($M=0.34$, $SD=0.22$) stroke ($t[82]=1.89$, $P=0.090$) was observed. No effect of task or strategy was observed in the stroke participants. Over 90% of stroke participants demonstrated competency on at least one of the 4 task-strategy combinations.

CONCLUSIONS: Children with perinatal stroke can achieve proficiency in basic tasks using simple BCI systems. Future directions include exploration of BCI-functional electrical stimulation systems for rehabilitation for children with hemiparesis and other forms of cerebral palsy.

GRAPHIC ABSTRACT: An online [graphic abstract](#) is available for this article.

Key Words: brain ■ brain computer interface ■ cerebral palsy ■ electroencephalography magnetic resonance imaging

Perinatal stroke is a focal, vascular brain injury occurring between 20 weeks gestation and the 28th postnatal day.¹ Perinatal stroke causes most hemiparetic cerebral palsy and lifelong physical disability for millions of persons worldwide.² Arterial ischemic strokes are the most common subtype and predominantly affect the middle cerebral artery, resulting in large combined cortical and subcortical lesions.^{3,4} In contrast, periventricular venous infarction occurs in the fetus before 34 weeks

gestation leading to subcortical infarction of the periventricular white matter.^{5–7} Most survivors of perinatal stroke suffer lifelong disabilities including sensorimotor deficits (cerebral palsy), disorders of language and learning, and epilepsy.² With limited understanding of pathophysiological mechanisms, prevention strategies are absent and the burden of perinatal stroke will persist.⁸

Perinatal stroke lesions often damage major components of the motor network, resulting in contralateral

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Nonstandard Abbreviations and Acronyms

BCI brain computer interface

hemiparetic cerebral palsy.² Children with severe hemiparesis often have little to no functional use of their affected hand and arm, resulting in difficulty carrying out activities of daily living including dressing, grooming, and feeding. Impairments often disrupt performance in school, limit recreational and employment opportunities, and decrease quality of life and self-esteem.^{9–11} Unfortunately, current rehabilitation options for motor dysfunction after perinatal stroke are limited. However, improved understanding of how the brain develops after early unilateral injury has produced sophisticated human models of developmental plasticity, including important roles for the contralesional hemisphere, which are affording new opportunities to improve function.¹²

A major aim of neurorehabilitation in children with cerebral palsy is to enhance their developmental trajectories so they might ultimately achieve higher levels of functioning and independence later in life.¹³ Upper extremity function is the most common target as it positively impacts activities of daily living and quality of life.¹¹ Existing rehabilitation options include physical therapy, occupational therapy, constraint induced movement therapy, and orthopedic surgery.¹⁴ While these methods may help, effect sizes are modest with variable response rates and those with the most severe deficits are even less likely to benefit. Recent trials of noninvasive brain stimulation suggest such applied neurotechnologies may further enhance neurorehabilitation in children with perinatal stroke, but efficacy remains to be proven and large doses of intensive therapy are required.^{12,15–17} There is therefore a pressing need to enhance the effectiveness of motor rehabilitation interventions for children disabled by perinatal stroke.

Brain computer interface (BCI) technology may provide novel opportunity to enhance rehabilitation of hemiparesis. BCI are a class of brain-machine interface with high potential to assist persons with neurological disabilities.^{18–20} BCI technology works by converting thought- or stimulation-induced alterations in brain activity into commands, which can then be used to control an external device.²¹ In this way, the user is able to affect their environment without engaging typical motor movement processes. Direct implantation of sensors into the brain has allowed severely disabled patients to gain remarkable control of effectors including robotic arms and computer systems.^{22,23} Noninvasive BCI alternatives via electroencephalogram (EEG) and other means are rapidly advancing.²⁴ Despite this potential, there is a general lack of awareness of BCI applications by clinicians caring for those with neurological disability.²⁵ Also

unfortunate is that nearly all BCI research has occurred in adults, neglecting the developing brains of children where clinical needs are high. We recently demonstrated that school-age children are highly capable of controlling simple, EEG-based BCI systems.²⁶ Applications in common pediatric disabilities like cerebral palsy remain underexplored.

Recent BCI applications for rehabilitation of adult stroke hemiparesis appear promising. Studies combining BCI with neuromuscular functional electrical stimulation (BCI-functional electrical stimulation) and visual feedback have suggested clinically important gains in upper extremity function.^{27,28} A sham-controlled, blinded, randomized clinical trial of 27 adults with moderate to severe upper extremity impairment secondary to stroke demonstrated positive effects of 10 hours of BCI-functional electrical stimulation that persisted for 6 to 12 months.²⁹ BCI coupled with physical therapy may also facilitate motor relearning after stroke.^{20,30} The potential for BCI approaches in perinatal stroke rehabilitation is supported by well understood models and clinical trial experience but is currently hampered by not knowing if or how such children can operate BCI systems.¹²

Here, we aimed to determine how well children with hemiparesis secondary to perinatal stroke can use a BCI to perform novel tasks using multiple strategies.

METHODS

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Participants

Participants were recruited from the Alberta Perinatal Stroke Project, a population-based cohort of >500 children with magnetic resonance imaging–confirmed perinatal stroke.³¹ Inclusion criteria were magnetic resonance imaging–confirmed perinatal stroke, clinically confirmed and disabling hemiparetic cerebral palsy, age 6 to 18 years, and informed consent. Children were excluded if they had neurological disorders not related to stroke, multifocal stroke, severe developmental delay and/or other inability to comply with study protocol, or unstable epilepsy. Typically developing control participants of the same age range were recruited from the community and completed identical procedures. Eligible families were contacted and recruited by a study team member and provided written informed consent and assent when appropriate. Informed consent was obtained for publication of any images. Methods were approved by the University of Calgary Research Ethics Board.

BCI System

We used the EMOTIV EPOC+, a commercially available, 14 channel, saline-based wet electrode, EEG headset (EMOTIV, San Francisco). The headset had preset electrode placement with regionally adjustable individual electrode positioning. Before use, electrodes were submerged in 0.9% saline. Electrodes were placed according to the international 10 to 20

system with sensors located at AF3, AF4, F3, F4, F7, F8, FC5, FC6, P7, P8, T7, T8, O1, O2. Two reference electrodes were placed on the left and right mastoids. Headset structure was fixed resulting in slight variability in location because of different participant head size and shape. EEG signal quality was confirmed via the EMOTIV Control Panel, the preprogrammed BCI software accompanying the system. All training and EEG recordings were completed using the same Control Panel and the Emotiv TestBench software, respectively.

EEG data was sampled at a rate of 2048 Hz, which was then down sampled to 128 Hz. Training consisted of 8-second EEG epochs. During these training periods, participants were instructed to perform their designated mental imagery strategy (see below). Event-related desynchronization of the EEG activity during this period was used to distinguish between resting and the trained strategy. Once the imagery strategy was detected, this was mapped to a keystroke command using Emokey (EMOTIV, San Francisco) and used to control one of 2 effector devices: a remote-controlled toy car via an Arduino interface circuit or a custom made computer cursor game.

BCI Strategies and Tasks

Participants completed 2 BCI sessions, 45 to 60 minutes in length, with each separated by ≈ 1 week. All sessions took place in a large, enclosed, quiet assessment room with reduced noise and distractions. During each session, participants completed both tasks and used one of 2 types of mental imagery.

Tasks included driving a remote controlled toy car and moving a cursor on a computer screen to a designated target. During the car task, the participant was comfortably seated with the toy car directly in front of them on the floor. A finish line was marked ≈ 4 feet away from the starting position. When instructed, the participant was first asked to remain neutral to keep the car still followed by using their designated imagery strategy to try and drive the car across the finish line. A Movie in the [Data Supplement](#) demonstrates a young child from pilot testing, not a study participant, wearing the EMOTIV headset and driving the toy car forward using a mental imagery strategy. When attempting the cursor task, the participant was seated in front of a computer monitor. A white cursor was positioned in the center of the screen with a red circle target to the right. The participant was instructed to maintain neutral at the beginning of each trial and then asked to use their imagination strategy to move the cursor toward the target. Movement of the device (car or cursor) was dependent only on the individual's mental imagery.

The 2 strategies were goal oriented (GO) and motor imagery (MI). During GO, the participant was instructed to think about pushing the effector device itself, either the car or cursor, toward the target. During MI, the participant was asked to remain still and simply imagine the opening and closing of both hands.

Participants were randomly assigned to their first task/strategy combination during session one. This then dictated the remaining combinations of task and strategy for the 2 sessions so that each contained the opposite combination of task and strategy. For example, if a participant began day 1 with car/GO their second task would be Cursor/MI. To account for order effects, the combinations were reversed for session 2 and the participant would begin with Cursor/GO and end with car/MI.

In this way, each participant attempted each task with a different strategy once. All possible task and strategy combinations across sessions are shown in Figure 1.

Training and Testing

Participants began each session with a 5-minute training period and then 10 trials of testing on 1 task, followed by a brief break and then another 5 minutes of training and 10 trials of testing on the second task. During training, participants alternated between remaining neutral and practicing a given imagery technique. Participants were first asked to remain neutral for 2 8-second blocks. Neutral was described as being relaxed, clearing one's thoughts or silently counting down backward from 10 to 1. This was followed by 2 8-second blocks of training the assigned imagery strategy, GO or MI. This was repeated 4 times to complete training for a single task. Neutral, GO, and MI attempts were labeled with manual markers during these recordings.

Training was immediately followed by 10 trials of testing. During the first 5 seconds of each trial, participants were asked to remain neutral. Participants were considered successful if they were able to keep the car or cursor stationary for the entire 5 second period. Next, the children were given 60 seconds to move the car or cursor toward its designated target. If able to produce any movement within the first 8 seconds, the attempt was recorded as successful.

The primary outcome was the agreement between the command given and the action produced as measured by Cohen kappa in keeping with similar previous BCI studies.^{32,33} All trials were video recorded to confirm timing and performance.

Tolerability

A tolerability survey consisting of questions related to participant fatigue, task and strategy preference, and comfort was administered at the end of each session. Questions were presented using Likert scale scoring (range from 1 to 5) and open text fields.

Analysis

Cohen kappa scores were tested for normality. Each participant had individual kappa scores for each task-strategy as well as a summary mean kappa score (the average of all task-strategy sessions). Age effects were explored using a Pearson correlation for scores across all task-strategy combinations. A 2-way repeated measures ANOVA was used to investigate effects of task or strategy on Cohen kappa. Independent

SESSION 1	SESSION 2
1. CAR MI → CURSOR GO	CURSOR MI → CAR GO
2. CAR GO → CURSOR MI	CURSOR GO → CAR MI
3. CURSOR MI → CAR GO	CAR MI → CURSOR GO
4. CURSOR GO → CAR MI	CAR GO → CURSOR MI

Figure 1. Tasks and imagery strategy groups.

All possible task and strategy combinations across both brain computer interface sessions. Each participant attempted each task with each strategy once between session 1 and 2. Order of tasks is counterbalanced between sessions.

samples *T*-tests were performed to compare average kappa scores for (1) controls versus perinatal stroke and (2) arterial versus venous strokes. We conducted a paired sample *T*-test to explore possible learning effects between session 1 and session 2 for all participants. Pearson correlation was performed to explore effect of strategy on trial-to-trial performance. χ^2 /Fisher exact tests were used to determine differences in preference between tasks and strategies. Statistical analysis was performed using SPSS 24, and graphs were created using SigmaPlot 12.5.

RESULTS

Demographics

We recruited 21 children with perinatal stroke, all of whom completed the study (57% male, mean age [SD] 13.5 [2.6] years, range 9–18). Stroke types were 11 arterial and 10 venous with no differences in demographics between groups. The majority of participants (15) had left side lesions and demonstrated left handedness. The study population demographics were comparable to 24 typically developing controls (71% male, mean age [SD] 13.7 [3.7] years, range 6–18). The majority of children (76%) reported using the BCI as a positive experience (4 or 5 on the 5-point scale). No serious adverse events occurred. No adverse side effects were reported although 62% of participants reported mental fatigue toward the end of their session.

BCI Performance

Cohen kappa scores for the entire sample were normally distributed with a mean kappa of 0.39 [0.18] and range of -0.10 to 1.00 . No correlation was observed between age and BCI performance ($r = -0.073$, $P = 0.51$, Figure 2). There was no difference in performance between males ($M = -0.42$, $SD = -0.27$) and females ($M = 0.35$, $SD = 0.25$) $t(82) = 1.28$, $P = 0.92$.

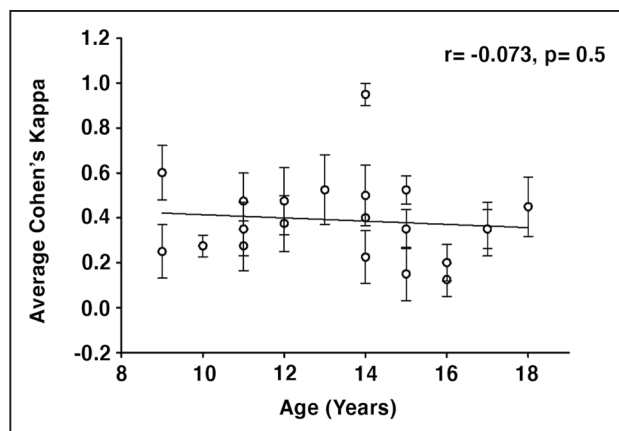


Figure 2. Average Cohen's Kappa score vs age.

No significant relationship between age and brain computer interface performance was observed (Pearson $r = -0.073$, $n = 21$, $P = 0.5$; error bars represent SEM).

Consistent with the adult literature, a kappa score of at least 0.4 was considered suggestive of BCI competency.^{33,34} When considering Cohen kappa scores across all task-strategy combinations, 9/21 (43%) of participants were able to meet the overall competency mark of 0.4 or higher. Almost all children (20/21, 95%) were able to achieve a Cohen's kappa of 0.40 or higher on at least one of the 4 the task-strategy combinations.

When comparing group BCI performance between perinatal stroke participants and typically developing controls, no difference in average Cohen's kappa scores were observed: stroke mean 0.39 (0.18) versus controls mean 0.42 (0.18) ($t(42) = 0.478$, $P = 0.94$, Figure 3A). Comparison of Cohen kappa scores between stroke types did not show a significant difference in performance between children with venous ($M = 0.45$, $SD = 0.29$) and arterial ($M = 0.34$, $SD = 0.22$) stroke $t(82) = 1.89$, $P = 0.090$ (Figure 3B).

Our previous study in healthy controls found effects of task and strategy on performance. Cohen kappa scores were significantly higher for the car task ($M = 0.52$, $SD = 0.30$) as compared with the cursor task ($M = 0.39$, $SD = 0.29$); $F(1, 25) = 5.52$, $P = 0.03$. Mean kappa scores for the GO strategy ($M = 0.51$, $SD = 0.32$) were significantly higher than for the MI ($M = 0.40$, $SD = 0.28$), $F(1, 25) = 5.20$, $P = 0.03$. This was in contrast to the perinatal stroke group where no effect of task or strategy was observed. The repeated measures ANOVA did not demonstrate a significant effect of task or strategy on BCI performance. There was no significant difference found between performance on the car ($M = 0.42$, $SD = 0.27$) and cursor tasks ($M = 0.35$, $SD = 0.25$); $t(82) = 1.05$, $P = 0.30$. Similarly, there was no difference in performance between the motor imagery ($M = 0.39$, $SD = 0.25$) versus goal oriented strategies ($M = 0.39$, $SD = 0.27$; Figure 4). The BCI experience survey suggested a slight preference for the GO strategy (57%) as compared with the MI (43%, $\chi^2 [2] = 0.43$, $P = 0.51$) and clear preference for the car task (81%) as compared with the cursor (19%, $\chi^2 [2] = 8.05$, $P = 0.01$).

No learning effects on BCI performance were observed between the first ($M = 0.40$, $SD = 0.26$) and second session in stroke participants ($M = 0.38$, $SD = 0.26$); $t(41) = 0.36$, $P = 0.72$ (Figure 5A). There did not seem to be any correlation between trial number and time to complete tasks for any of the 4 groups (Figure 5B). There was also no correlation between trial and performance on goal oriented ($r = 0.024$, $n = 420$, $P = 0.63$) or motor imagery-based tasks ($r = -0.050$, $n = 420$, $P = 0.30$; Figure 5C). Comparisons of BCI performance between lesion sides demonstrated that participants with right side lesions ($M = 0.41$, $SD = 0.33$) had significantly higher Cohen kappa scores as compared with those with left lesions ($M = 0.38$, $SD = 0.23$); $F(1, 82) = 4.28$, $P = 0.04$.

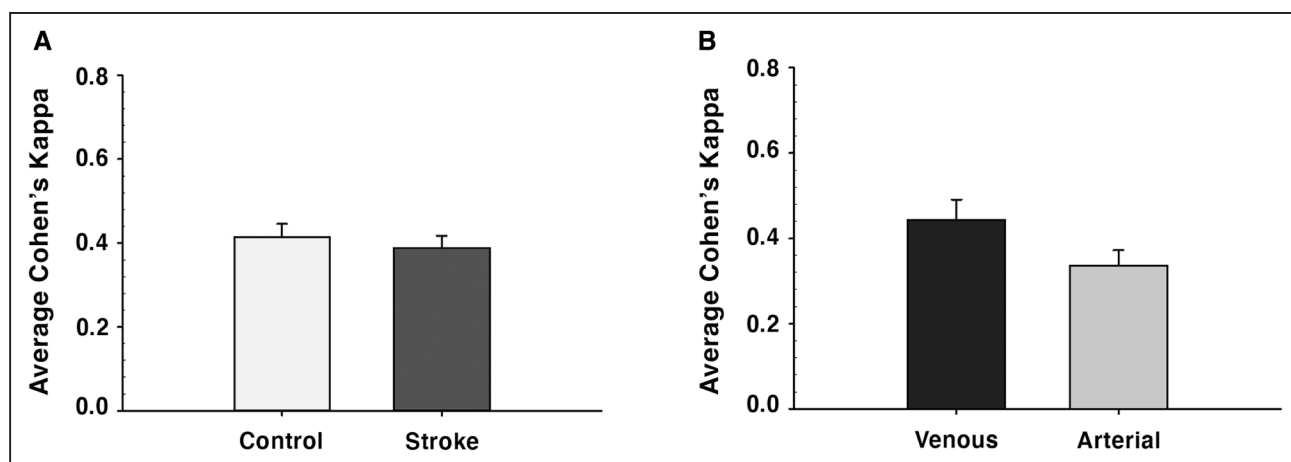


Figure 3. Mean Cohen's Kappa score across controls and stroke types.

A, Mean Cohen's Kappa for typically developing children ($M=0.42$, $SD=0.18$) compared with children with perinatal stroke ($M=0.39$, $SD=0.18$) and **(B)** children with venous stroke ($M=0.45$, $SD=0.29$) vs arterial stroke ($M=0.34$, $SD=0.22$). No significant differences were observed between groups.

DISCUSSION

Our study suggests that, despite early and often large brain injuries, children with perinatal stroke can use a simple BCI system with similar competency to typically developing children. We did not find a clear effects of age, task, strategy, or stroke type on performance or strong learning effects either within or between sessions. Our results support the potential use of BCI technology in this clinical population. While each child's individual performance varied across tasks and strategies, almost all participants were able to achieve competency on at least 1 task/strategy combination. BCI performance is highly individualized, and these results suggest that training may be tailored to each individual child.

While we observed that children with perinatal stroke can use a simple BCI, our results suggest some differences in performance as compared with their typically developing peers. For example, our previous study in

healthy controls demonstrated age effects, with older children performing better on BCI tasks.²⁶ Typically developing children also showed better performance during GO imagery and the car task as compared with the MI strategy and cursor task. Similar patterns in performance were not observed in our perinatal stroke group despite comparable power.²⁴ BCI studies in children, particularly in those with neurological disability, have been rare, and we do not yet understand how the mechanisms of control differ between controls and such pediatric clinical populations.

Differences are likely in part attributable to the increasingly understood alterations that occur in motor system organization after early unilateral brain injury.^{12,35,36} Following perinatal stroke, injured upper motor neurons have a decreased ability to compete for synapses in the anterior horn, resulting in increased success and persistence of ipsilateral projections normally present at birth.¹⁵ This arrangement of bilateral motor control may also have implications for the organization of the motor imagination used to drive BCI paradigms. Abnormalities of the EEG itself are also common in this population, even in the absence of epilepsy.³⁷ Variations in EEG activity likely impact multiple factors important for BCI performance including location and dynamics of sensorimotor rhythms, event-related desynchronizations, and potential performance factors such as attention. Use of higher quality and density EEG in future studies may both inform mechanisms and improve performance in this population.

We had initially hypothesized that children with venous strokes would have significantly better performance as compared with children with arterial infarcts. This was largely due to the fact that arterial strokes often result in damage to cortical regions implicated in mental imagery. Studies in healthy adults have suggested that the presupplementary motor area, dorsal premotor cortex, superior parietal lobe, are important cortical areas used

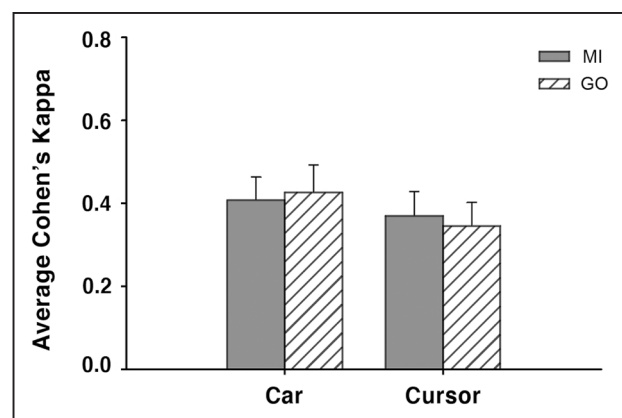


Figure 4. Effect of task and strategy on brain computer interface performance.

No significant effect was observed for task/strategy on Cohen's Kappa scores (Mean Cohen's Kappa for car [$M=0.42$, $SD=0.27$], cursor [$M=0.35$, $SD=0.25$], motor imagery [MI; $M=0.39$, $SD=0.25$], and goal-oriented [GO] strategies [$M=0.39$, $SD=0.27$]).

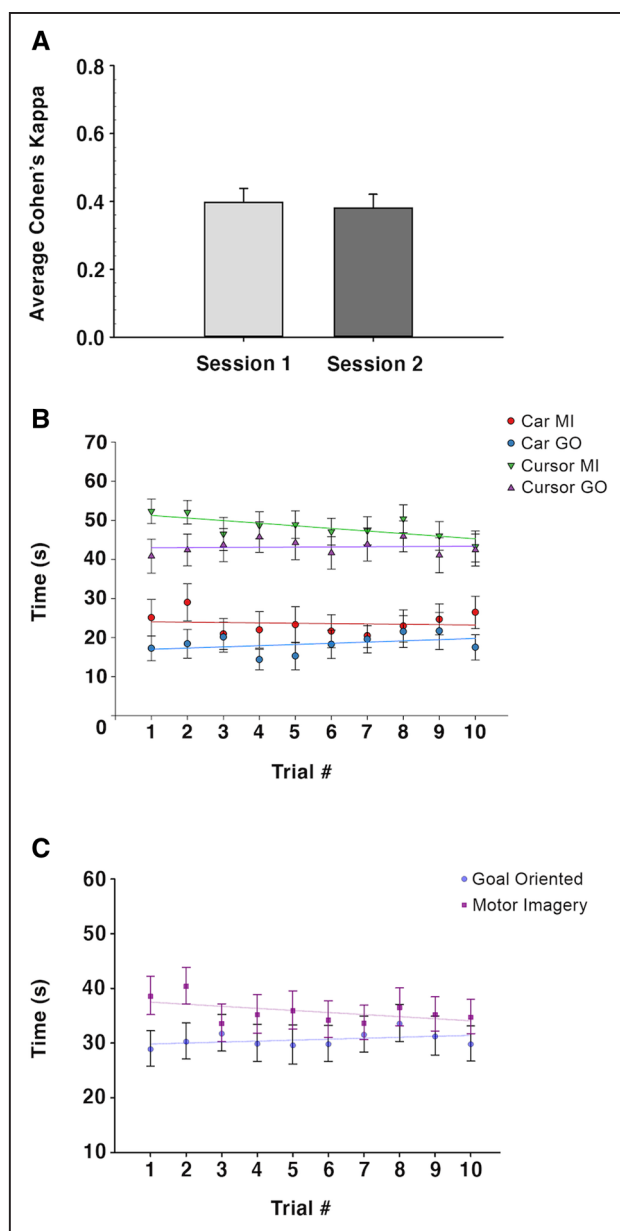


Figure 5. Effects of session and trial number on brain computer interface performance and timing.

A, Mean Cohen's Kappa for session 1 ($M=0.40$, $SD=0.26$) and session 2 ($M=0.38$, $SD=0.26$). No significant difference in performance between sessions, **(B)** relationship between time to complete trial and task/strategy combination, **(C)** relationship between time to complete trial across all goal oriented ($r=0.024$, $n=420$, $P=0.63$) and motor imagery attempts ($r=-0.050$, $n=420$, $P=0.30$).

during mental imagination of a movement.³⁸ However, we did not observe major performance differences between stroke groups, which may in part be related to the often dramatic developmental cortical reorganization known to occur in this population.

Although there was no observed difference between performance on the 2 tasks, the majority of participants expressed a preference for the car task. This was perhaps not surprising as the computer cursor task was quite boring in comparison. Utilizing more engaging

tasks for BCI training may be beneficial for sustaining participant interest over longer periods of time and is particularly relevant if using BCI in a pediatric population. We also did not observe any strong effect of trial or session number on performance as has been observed previously in healthy children. This may have been due to mental fatigue or frustration as the session progressed or difficulty reproducing the consistent mental imagery during the testing period. User mental state has previously been linked to BCI performance in healthy adults. An optimal level of mental fatigue and frustration may be beneficial for BCI performance, and more research is required to define these issues in children.³⁹

Our results suggested higher BCI performance in participants with right-sided lesions (right handed). These results are supported by a recent study demonstrating poorer BCI performance in healthy, left handed, adults.⁴⁰ While this may indicate an underlying difference in mechanism for BCI control related to handedness and lesion side, we are cautious to draw these conclusions in our population at this point. This is in part due to the difference between the number of participants with left- and right-sided lesions as well as the further differences in stroke type (venous versus arterial) and lesion size between participants. Further studies may consider a more standardized approach to exploring effect lesion side on BCI performance.

Our results have implications for clinical applications of BCI systems in pediatric populations, specifically children with early brain injuries and cerebral palsy. Having demonstrated that children with unilateral perinatal brain injury are able to use a simple BCI system opens additional opportunities to explore the potential of BCI in pediatric neurorehabilitation. For example, BCI-activated functional electrical stimulation has shown significant improvements in upper extremity function in adults and could be an impactful approach for children with hemiparesis.^{29,41} These opportunities are not isolated to the perinatal stroke population. Our findings lend support to the use of BCI technology in other clinical populations including more severe quadriplegic cerebral palsy. BCI has the potential to allow such children with extremely limited mobility and communication new opportunities for communication and interaction with their environment.

Working with simple BCI tools also allows us to better understand the fundamentals of how this technology might work differently in children. Although integration of BCI technology into rehabilitation paradigms would likely require more robust systems and training, these preliminary steps are encouraging. Continued exploration of how BCI is tolerated and the neural mechanisms behind BCI control in children brings the potential to create novel and needed rehabilitation tools for those who need them most. Significant limitations remain. We were certainly limited by the EMOTIV EPOC+ headset design, required to work with a fixed

number of electrodes in relatively immovable positions. It is likely that for some participants we were recording EEG over large infarcts and whether this is just an absence of signal or adds confounding noise to the system needs to be determined. Future studies using a modifiable EEG cap would be beneficial in personalizing the BCI session for each participant. Future efforts to study BCI in children with perinatal stroke also need to consider differing mechanisms of injury and how neural development and activity have been altered. That these and other factors almost certainly vary across individuals needs to be considered in study design if personalized approaches are to rehabilitation are to be achieved.

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Dr Jadavji conceptualized and designed the study, collected the data, conducted analyses, and drafted and revised the initial article. J. Zhang conceptualized and designed the study and revised the article. B. Paffrath helped design the BCI car control device. Drs Zewdie and Kirton conceptualized and designed the study and reviewed and revised the article.

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Disclosures

None.

Supplemental Materials

Online Movie

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